

Time Series Exercise Sheet 10

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Exercise 10.1

Let ε_t be a white noise process. Show that the best one-step-ahead predictors for the causal $AR(2)$ process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$$

based on one observation X_1 and on two observations X_1, X_2 are

$$X_2^1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1, \quad X_3^2 = \phi_1 X_2 + \phi_2 X_1.$$

Solution 10.1

For the one-step-ahead best linear predictor based on X_1 , we use the theorem on prediction equations seen in the course, we get $\gamma_0 \beta_1 = \gamma_1$, which leads to

$$X_2^1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1.$$

For the one-step-ahead best linear predictor based on X_1, X_2 , we could try to reuse the theorem and build the matrix $\mathbf{\Gamma}_n$. Instead, we notice that $X_3 = \phi_1 X_2 + \phi_2 X_1 + \varepsilon_3$ implies that $X_3 - \phi_1 X_2 + \phi_2 X_1 = \varepsilon_3$. Because ε_t is a white noise process, we get

$$\begin{aligned} \mathbb{E}[(X_3 - \phi_1 X_2 + \phi_2 X_1) X_1] &= \mathbb{E}[\varepsilon_3 X_1] = 0 \\ \mathbb{E}[(X_3 - \phi_1 X_2 + \phi_2 X_1) X_2] &= \mathbb{E}[\varepsilon_3 X_2] = 0, \end{aligned}$$

which implies that $X_3^2 = \phi_1 X_2 + \phi_2 X_1$.

Exercise 10.2

Theorem. For a stationary process $\{X_t\}$, X_{n+h}^n is found by solving for $\beta_0, \dots, \beta_n$ the prediction equations

$$\mathbb{E}[(X_{n+h} - X_{n+h}^n) X_k] = 0, \quad k = 0, \dots, n,$$

where $X_0 = 1$, and $X_{n+h}^n = \beta_0 + \sum_{j=1}^n \beta_j X_j$. Let $\mu = \mathbb{E}[X_t]$.

The β_j are then given by the solution of the system of equations

$$\mathbf{\Gamma}_n \boldsymbol{\beta} = \boldsymbol{\gamma}_{[h]}, \quad \beta_0 = \mu(1 - \boldsymbol{\beta}^T \mathbf{1}_n)$$

with

$$\mathbf{\Gamma}_n = \begin{pmatrix} \gamma_0 & \gamma_1 & \dots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & \dots & \gamma_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{n-1} & \gamma_{n-2} & \dots & \gamma_0 \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix}, \text{ and } \boldsymbol{\gamma}_{[h]} = \begin{pmatrix} \gamma_{n+h-1} \\ \gamma_{n+h-2} \\ \vdots \\ \gamma_h \end{pmatrix}. \quad (1)$$

1. Prove the above theorem

(a) Express the prediction mean square error of X_{n+h}^n in matrix form in terms of $\text{Var}(\mathbf{X})$, $\boldsymbol{\beta}$ and β_0 .

- (b) Find the expression of β_0 by using the properties of the best linear predictor.
- (c) Explain why there is no loss of generality in considering $\mu = 0$. What is then the value of β_0 ?
- (d) Show that if $\mu = 0$, then the best linear predictor satisfies

$$\text{Var}(\mathbf{X})\boldsymbol{\beta} = \text{Cov}(X_{n+h}, \mathbf{X}). \quad (2)$$

- (e) Show that the prediction equations for $k = 1, \dots, n$ are equivalent to (2). Explain how you found the prediction equation for $k = 0$ in the previous steps.

2. Assume $\mathbf{\Gamma}_n$ is invertible.

- (a) Express X_{n+h}^n in matrix form. How does it differ from the result seen in the lecture (e.g. $\mu = 0$)?
- (b) What is the prediction mean squared error? How does it differ from the result seen in the lecture?

Solution 10.2

1. (a) The prediction mean square error

$$\begin{aligned} S(\beta_0, \dots, \beta_n) &= \mathbb{E} \left[\left(X_{n+h} - \beta_0 - \sum_{j=1}^n \beta_j X_j \right)^2 \right] \\ &= \mathbb{E} \left[(X_{n+h} - \beta_0 - \boldsymbol{\beta}^T \mathbf{X})^2 \right] \\ &= \text{Var}(X_{n+h} - \beta_0 - \boldsymbol{\beta}^T \mathbf{X}) + \mathbb{E} [X_{n+h} - \beta_0 - \boldsymbol{\beta}^T \mathbf{X}]^2 \\ &= \gamma_0 - 2\boldsymbol{\beta}^T \text{Cov}(X_{n+h}, \mathbf{X}) + \boldsymbol{\beta}^T \text{Var}(\mathbf{X})\boldsymbol{\beta} + (\mu - \beta_0 - \mu\boldsymbol{\beta}^T \mathbf{1}_n)^2 \end{aligned}$$

is a quadratic function bounded below by zero, so it certainly has a minimum that satisfies the equations $\partial S / \partial \beta_r = 0$ for $r = 0, \dots, n$.

- (b) The first equation $\partial S / \partial \beta_0 = 0$ gives

$$\frac{\partial S}{\partial \beta_0} = 2\beta_0 - 2(\mu - \mu\boldsymbol{\beta}^T \mathbf{1}_n) = 0 \implies \beta_0 = \mu(1 - \boldsymbol{\beta}^T \mathbf{1}_n).$$

- (c) Hence the form of the best linear predictor is $X_{n+h}^n = \beta_0 + \boldsymbol{\beta}^T \mathbf{X} = \mu + \boldsymbol{\beta}^T (\mathbf{X} - \mu \mathbf{1}_n)$. Thus there is no loss of generality in considering $\mu = 0$ in which case $\beta_0 = 0$ and $X_{n+h}^n = \boldsymbol{\beta}^T \mathbf{X}$.
- (d) When $\mu = 0$,

$$S(\boldsymbol{\beta}) = \gamma_0 - 2\boldsymbol{\beta}^T \text{cov}(X_{n+h}, \mathbf{X}) + \boldsymbol{\beta}^T \text{var}(\mathbf{X})\boldsymbol{\beta}.$$

We differentiate $S(\boldsymbol{\beta})$ to find the matrix equation $\text{var}(\mathbf{X})\boldsymbol{\beta} = \text{cov}(X_{n+h}, \mathbf{X})$ as stated in the theorem.

- (e) This matrix equation may be rewritten as the prediction equations, the first of which, for β_0 , gives $\mu - \beta_0 - \sum_{j=1}^n \beta_j \mu = 0$, and hence the required form for β_0 . The other equations for $k = 1, \dots, n$ are obtained by noting that

$$\mathbb{E} \left[\left(X_{n+h} - \sum_{j=1}^n \beta_j X_j \right) X_k \right] = 0 \iff \gamma_{n+h-k} = \sum_{j=1}^n \beta_j \gamma_{jk}, \quad k = 1, \dots, n,$$

which is equivalent to the matrix equation (write $\text{Var}(\mathbf{X})\boldsymbol{\beta} = \text{Cov}(X_{n+h}, \mathbf{X})$ and identify each row equation).

2. (a) If $\mathbf{\Gamma}_n$ is invertible, then we have $\boldsymbol{\beta} = \mathbf{\Gamma}_n^{-1} \boldsymbol{\gamma}_{[h]}$, which leads to

$$X_{n+h}^n = \mu + \boldsymbol{\gamma}_{[h]}^T \mathbf{\Gamma}_n^{-1} (\mathbf{X} - \mu \mathbf{1}_n).$$

(b) Since

$$\mathbb{E} [X_{n+h} - X_{n+h}^n] = 0,$$

the prediction mean square error is

$$\text{Var} (X_{n+h} - X_{n+h}^n) = \text{Var} (X_{n+h} - \beta^T \mathbf{X}) = \gamma_0 - 2\beta^T \boldsymbol{\gamma}_{[h]} + \beta^T \boldsymbol{\Gamma}_n \beta = \gamma_0 - \boldsymbol{\gamma}^h{}^T \boldsymbol{\Gamma}_n^{-1} \boldsymbol{\gamma}_{[h]},$$

which is the same as in the zero-mean case.

Exercise 10.3

Prove the following theorem

Theorem. *The best linear predictor X_{n+h}^n for X_{n+h} in a causal ARMA process with general linear representation $\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ is*

$$X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j} = \psi_h \varepsilon_n + \psi_{h+1} \varepsilon_{n-1} + \cdots.$$

The corresponding prediction mean square error is $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$.

Solution 10.3

We consider the linear predictor $X_{n+h}^n = \sum_{i=1}^n \beta_i X_i$, and on noting the linear representation $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$, we see that we can write

$$X_{n+h}^n = \sum_{i=1}^n \beta_i \sum_{j=0}^{\infty} \psi_j \varepsilon_{i-j} = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j},$$

say, i.e., X_{n+h}^n has a linear representation starting at ε_n . The prediction mean square error,

$$\mathbb{E} \left[(X_{n+h} - X_{n+h}^n)^2 \right] = \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j} - \sum_{j=0}^{\infty} c_j \varepsilon_{n-j} \right)^2 \right] = \sigma^2 \left\{ \sum_{j=0}^{h-1} \psi_j^2 + \sum_{j=h}^{\infty} (\psi_j - c_{j-h})^2 \right\}$$

is minimised by taking $c_j = \psi_{j+h}$, for $j = 0, 1, \dots$, and the prediction is then

$$X_{n+h}^n = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j} = \sum_{j=0}^{\infty} \psi_{j+h} \varepsilon_{n-j} = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j}.$$

Since $X_{n+h} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j}$, we see that $X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}$, and the mean square error is

$$P_{n+h}^n = \mathbb{E} \left\{ (X_{n+h} - X_{n+h}^n)^2 \right\} = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$